Technical assessment

1 Context

Real-estate projects slow down when the first regulatory study is manual, fragmented, and hard to audit. A useful AI system for this workflow should not merely answer isolated zoning questions: it should identify the applicable rules, trace every conclusion back to the source documents, and synthesize those rules into a practical development output.

This take-home focuses on exactly that problem.

2 Assessment

This is a **48-hour take-home assessment**.

Your task is to build an **agent** — using the framework of your choice — that can automatically navigate the attached planning documents and produce a **full regulatory analysis** for the parcel located at:

4 rue Joliot Curie, 91190 Gif-sur-Yvette (Essonne)

The target program is:

- **one Résidence Sociale Étudiante (RSE)**, and
- **one Résidence Jeunes Actifs**,

in **two separate buildings**.

The central output we want is the parcel's **buildable envelope**: in other words, the **authorized 3D volume** inside which construction is allowed under the attached regulations.

A **3D visualization** of that volume is welcome, but it is **not mandatory**. The bare minimum is a working agent that performs a serious regulatory analysis for this parcel and produces a justified buildability study.

The assessment has two parts: building the agent (Part 1) and designing the evals that measure whether it is correct (Part 2). **Both parts are required.** A submission without serious evals is incomplete, however good the agent's output looks, so plan your time accordingly.

3 Authoritative inputs

Your agent must automatically work through the **three attached regulatory documents**:

- `plu_reglement_91272_20210706.pdf` — PLU regulatory text,
- `plu_reglement_graphique_91272_20210706.pdf` — PLU graphic regulation document,
- `oap_91272_20210706.pdf` — OAP document, including a chapter on *Le quartier Joliot Curie*.

Your system should determine which parts of these documents apply to the parcel and the proposed program. In particular, it should not assume applicability in advance; it should **identify and justify** the relevant zone, overlays, graphic prescriptions, and OAP provisions.

You may use additional **public, non-regulatory data sources** when needed to identify the parcel geometry or locate the address on a map (for example cadastral or basemap data), but you must clearly separate:

- **regulatory sources**, which justify legal/planning conclusions, from
- **geometric/contextual sources**, which help locate and represent the parcel.

4 Part 1: Build a regulatory-analysis agent

Build an agent that can ingest, navigate, retrieve from, and reason over the attached documents in order to produce a parcel-specific development study.

At minimum, your agent should do the following:

- identify the parcel corresponding to **4 rue Joliot Curie, 91190 Gif-sur-Yvette**,
- determine the **applicable regulatory regime** for that parcel,
- determine whether the target program — **one RSE** and **one Résidence Jeunes Actifs** in **two separate buildings** — is permitted, restricted, conditional, or incompatible under the applicable documents, and explain how you map that program to the relevant regulatory use / destination categories when needed,
- extract and interpret the rules that shape the parcel's **buildable envelope**,
- synthesize those rules into a **clear buildability conclusion**,
- and cite the supporting sources precisely enough that a human reviewer can audit every important conclusion.

4.1 What we mean by "buildable envelope"

For this assessment, the buildable envelope is the maximum admissible **3D construction volume** allowed by the applicable rules, based on the documents provided.

Depending on what the documents say, this may require reasoning over items such as:

- land-use / destination admissibility,
- alignments and mandatory build-to lines,
- setbacks from parcel limits or public space,
- distances between buildings,
- height limits, step-backs, or prospect rules,
- footprint or ground-coverage constraints,
- protected strips, landscape constraints, reserved spaces, or graphic prescriptions,
- OAP-level urban design prescriptions,
- and any other rule in the attached documents that materially affects what can be built.

If some rules affect **overall feasibility** rather than the envelope directly — for example parking, access, servicing, or other conditions — you should still surface them clearly in the final study.

4.2 Expected output of Part 1

Your final parcel study should be specific, auditable, and useful to a development team. At minimum, it should contain:

- the parcel identification method and any geometry assumptions,
- the applicable zoning / overlays / OAP provisions,
- a reasoned statement on the admissibility of the two-building program,
- the rules that govern the buildable envelope,
- the resulting envelope conclusion,
- all major assumptions, ambiguities, and unresolved points,
- and source citations to the attached documents.

A 2D diagram or 3D massing visualization is a plus, but the minimum acceptable deliverable is a rigorous text-and-calculation based regulatory analysis.

5 Part 2: Evaluation of the agent

This part is **mandatory** and is graded with the same weight as system design. We want to see how you prove that your agent is correct, and we expect you to spend a substantial share of your time here.

You must design a set of **carefully crafted evals** to measure whether your agent is actually correct.

Your eval section must answer all of the following:

- **What exactly did you evaluate?**
- **How did you build the eval set(s)?**
- **Why are these evals difficult to game?**
- **What score did your agent achieve?**
- **What kinds of failures did the evals reveal?**

You are free to choose the exact eval design, but it should be **serious**, **well-justified**, and **hard to game**.

5.1 Minimum evaluation requirements

At minimum, your submission must include:

- at least one **explicit eval set** with clearly defined expected answers,
- a **scoring method** that you explain,
- the agent's **score(s)** on that eval set,
- and a short discussion of **failure modes**.

If your evals contain numerical answers, define your tolerance rules. If they contain citations, define how citation correctness is judged. If they contain multi-step reasoning, explain how you distinguish a lucky final answer from grounded reasoning.

6 System-design expectations

We are deliberately leaving the implementation open. You may use the framework, model stack, retrieval approach, orchestration approach, or memory structure of your choice.

7 How we evaluate submissions

We will evaluate submissions primarily on three dimensions:

- **System design (40%).** We will look at the quality of the architecture, the thoughtfulness of the workflow, the handling of evidence and calculations, and the overall soundness of the agentic approach.
- **Eval design and agent performance on the evals (40%).** We will look first at the quality of the evals themselves, then at the agent's score on them. A simple benchmark with inflated scores will not be rewarded as highly as a serious benchmark that exposes real strengths and weaknesses.
- **Final study correctness and quality (20%).** We care about whether the produced regulatory analysis is convincing, well cited, and useful in practice.

In other words: the best submission is not necessarily the one with the flashiest demo, but the one that shows disciplined system design, credible evaluation, and a trustworthy parcel study.